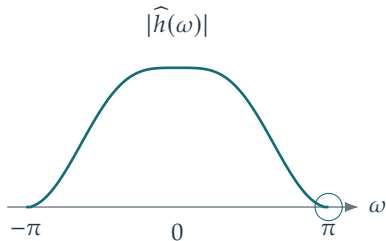
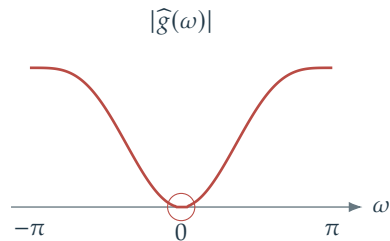


two vanishing moments illustrated ($p = 2$)



order- p zero at π



order- p zero at 0

$$\widehat{h}^{(k)}(\pi) = \widehat{g}^{(k)}(0) = 0 \iff \int_{\mathbb{R}} x^k \psi(x) \, dx = 0, \quad 0 \leq k < p$$

Curves show magnitudes; derivatives refer to the filter symbols.